

Quiz — 2.1.4 Thinking Logically

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 · Component 02 · 2.1.4

Section A — Multiple choice (6 marks)

1. In a program, a *decision point* is a place where: A. Data is stored in fast memory for reuse B. The flow of control branches depending on a condition C. A problem is broken into sub-procedures D. Two tasks run at the same time Your answer: ☐
2. A cinema offers a discount if the customer is under 16. The correct Boolean condition for the decision point is: A. `IF age = 16` B. `IF age < 16` C. `IF age >= 16` D. `IF age > 16` Your answer: ☐
3. A booking system confirms a seat only when `seatsRequested <= seatsAvailable`. What happens on the **FALSE** branch of this decision? A. The seats are booked successfully B. The booking is rejected / the user is told there are too few seats C. The seats are cached D. Nothing — there is no false branch Your answer: ☐
4. A thermostat turns the heating ON when the temperature is below 18°C, otherwise OFF. How many outcomes (branches) does this single decision point have? A. None B. One C. Two D. Four Your answer: ☐
5. A loop "repeat while there are more records to process" relies on a condition tested each iteration. This loop test is an example of: A. A reusable component B. A decision point controlling flow of control C. Abstraction D. A precondition that is only checked once Your answer: ☐
6. A login system locks the account after 3 failed attempts. The decision should fire when: A. `IF attempts = 1` B. `IF attempts >= 3` C. `IF attempts < 3` D. `IF attempts <= 0` Your answer: ☐
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Section B — Short answer (9 marks)

7. A game awards a bonus life when a player's score reaches 1000 or more. Write the exact Boolean **condition** for this decision point and state the outcome of **both** branches. [3]
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8. A car park barrier opens only if a ticket is valid **and** the fee has been paid. Write a suitable Boolean condition using a logical operator, and state what happens when the condition is false. [2]
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9. Explain what is meant by *flow of control* and how decision points affect it, using a simple example. [2]
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10. A shop offers free delivery when the basket total is over £50. A student writes `IF total > 50`. State whether a £50 order qualifies under this condition, and rewrite the condition if free delivery should include

exactly £50. [2]

Section C — Applied question (5 marks)

11. A streaming service decides which version of a film to play. If the user has a premium subscription it streams in 4K; otherwise it streams in HD, unless the user's connection is slow (below 5 Mbps), in which case it always streams in SD regardless of subscription. Using *thinking logically*, identify the decision points, write exact Boolean conditions for each, and state the outcome of each branch. [5]

Answer key

Section A (6 marks — 1 each) 1. **B** — a decision point is where flow of control branches on a condition. 2. **B** — "under 16" means strictly less than 16, so `age < 16`. 3. **B** — when the condition is false there are not enough seats, so the booking is rejected. 4. **C** — a single IF/ELSE decision has two outcomes (ON and OFF). 5. **B** — a loop's repeat-condition is a decision point tested each pass, steering flow of control. 6. **B** — locking on the third failure means `attempts >= 3`.

Section B (9 marks)

7. [3] Condition (1): `IF score >= 1000`. TRUE branch (1): award the player a bonus life. FALSE branch (1): do not award a life / continue play unchanged. Must give the exact `>=` Boolean *and* both outcomes.
8. [2] Condition (1): `IF ticketValid = TRUE AND feePaid = TRUE` (accept `IF ticketValid AND feePaid`). False outcome (1): the barrier stays closed / the driver is refused exit and prompted to pay. Must use a logical operator (AND).
9. [2] Flow of control = the order in which a program's statements are executed (1). Decision points alter that order by branching to different statements depending on whether a condition is true or false — e.g. `IF balance >= price` runs the "buy" path, otherwise the "decline" path (1).
10. [2] Under `total > 50` a £50 order does **not** qualify, because 50 is not greater than 50 (1). To include exactly £50, use `IF total >= 50` (1).

Section C (5 marks) 11. [5] Up to 5 from: First/overriding decision on connection speed — `IF connectionSpeed < 5` (Mbps) THEN stream SD, regardless of subscription (1 for condition, 1 for both/overriding outcome). Otherwise (connection adequate) decide on subscription — `IF subscription = "premium"` (or `isPremium = TRUE`) THEN stream 4K, ELSE stream HD (1 for condition, 1 for both branch outcomes). Recognise the conditions are nested/ordered so the speed test takes priority (1). Conditions must be exact Booleans with both branches stated and applied to this scenario; vague "if it's fast" wording does not earn the condition marks.

Total: 6 + 9 + 5 = 20